

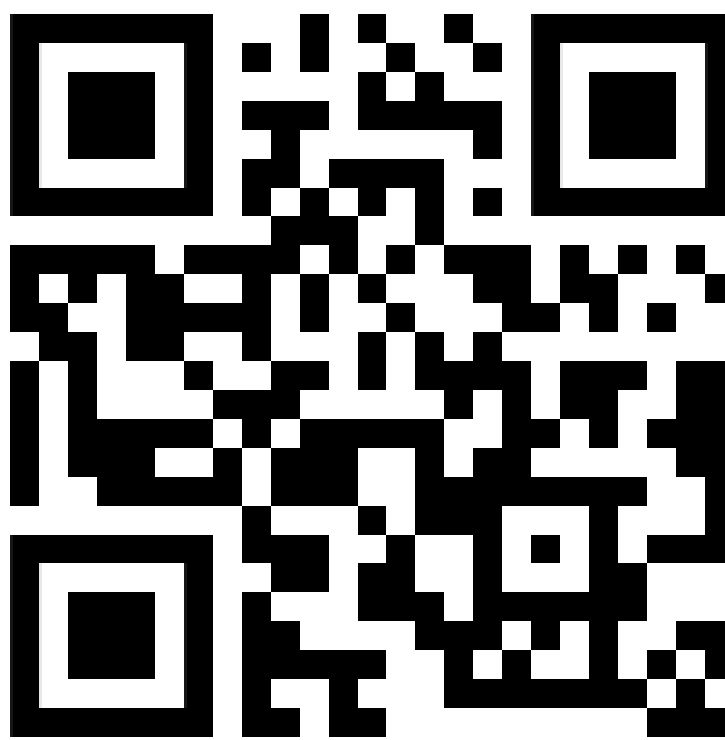
Introduction

- **Scalar implicatures (SIs)** are commonly studied using **sentence verification tasks**.
- Linking assumption: Computing an SI  $\Rightarrow$  Rejecting an underinformative sentence (e.g., “Some trees are plants.”)
- Rates of pragmatic (rejecting) vs. literal (accepting) responses vary a lot across studies. [1, 2, 3, 4, 5, 6]
- **Hypothesis:** Literal response rates depend on question framing (truth vs. felicity) and context source (world knowledge vs. picture).

The Present Study

- We ran four web-based experiments (252, 252, 576, and 576 participants) on Prolific with PClbex. [7]
- Each experiment employed a sentence(–picture) verification task. [8, 9]
- We manipulated whether truth (true/false) or felicity (good/bad) judgments were asked for. [10]
- We also manipulated whether sentences were judged against world knowledge (WK) or a picture.
- Exps. 1–3 were run in English; Exp. 4 replicated Exp. 3 in French (*certaines/certaines*).
- Results show higher literal response rates for truth (over felicity) and world knowledge (over pictures).

Data & Code:



osf.io/86r4u

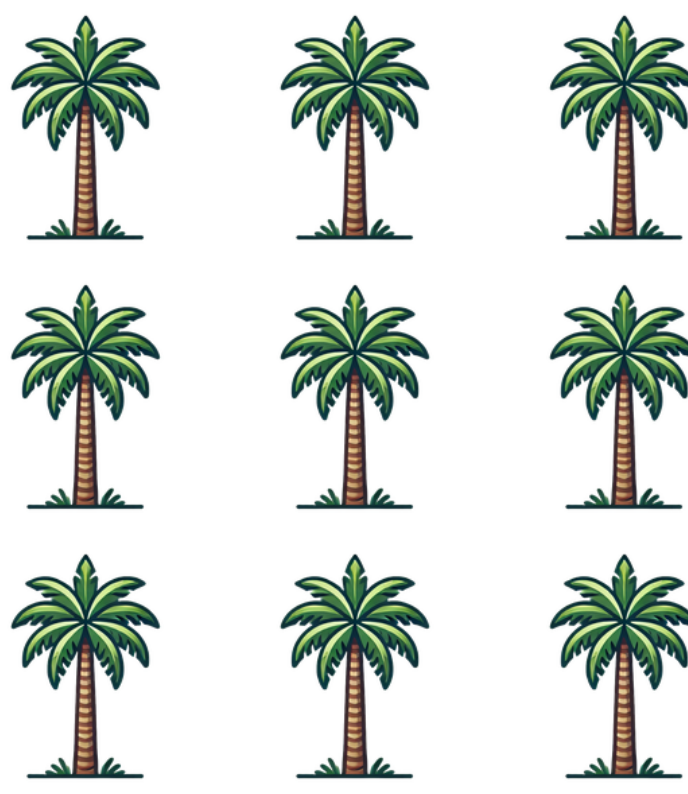
Experimental Design

| Group | Sentence                     | Question                                              | Question Framing | Context Source  |
|-------|------------------------------|-------------------------------------------------------|------------------|-----------------|
| a     | Some trees are plants.       | Is this a true or a false statement?                  | Truth            | World knowledge |
| b     | Some of the trees are palms. | Is this a true or a false description of the picture? | Truth            | Picture         |
| c     | Some trees are plants.       | Is this a good or a bad statement?                    | Felicity         | World knowledge |
| d     | Some of the trees are palms. | Is this a good or a bad description of the picture?   | Felicity         | Picture         |

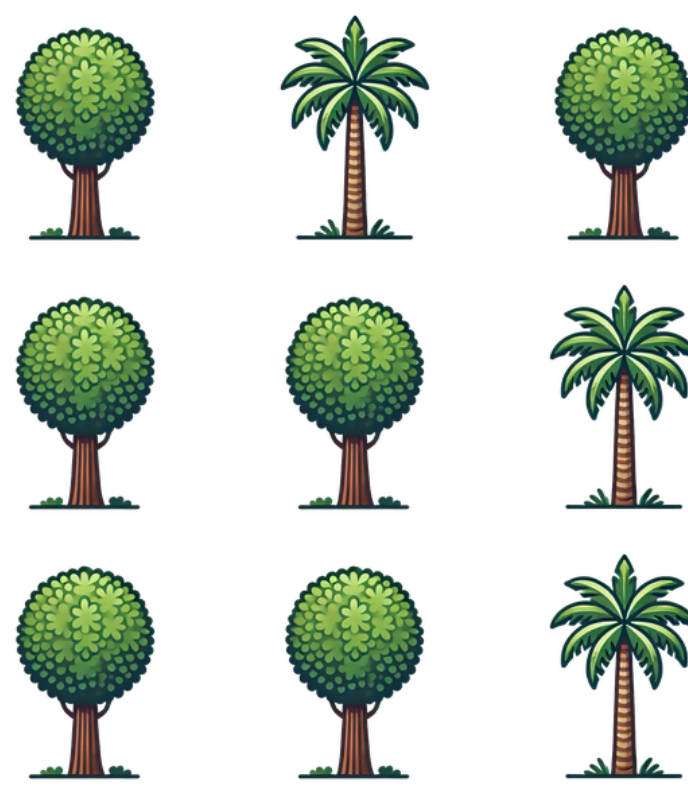
- Between-subjects design (groups a, b, c, d); 24 items

Differences Between Experiments

- Exp. 1:** 3 trial types (target, control-true, control-false)
- Exp. 2:** like Exp. 1, but question “Is this a true statement?”
- Exp. 3:** 6 trial types (3 new controls with strong scalemate)
- Exp. 4:** like Exp. 3, but in French (scalar *certaines/certaines*)



Target

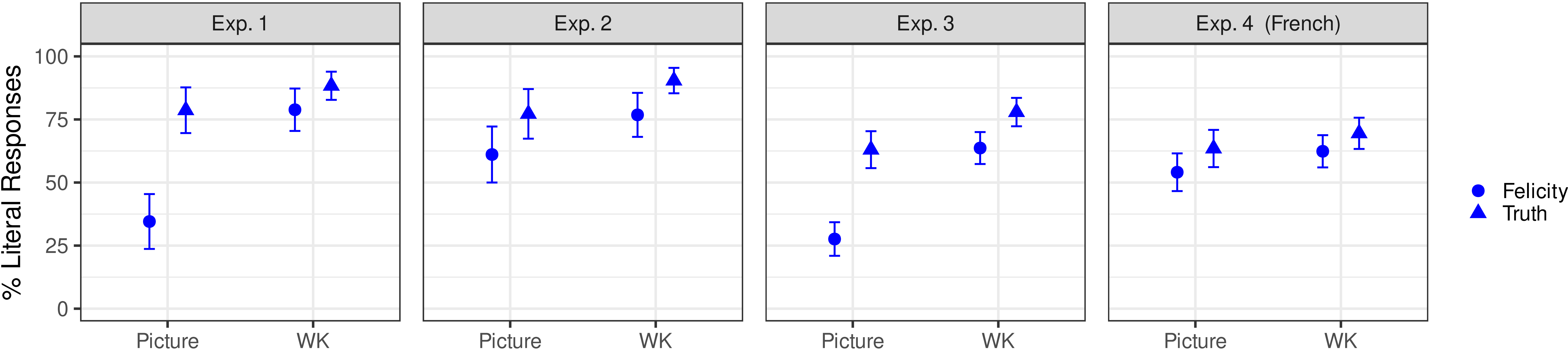


Control true/good



Control false/bad

Results



| Model estimates  | Truth vs. felicity |           |          |          | WK vs. picture |           |          |          |
|------------------|--------------------|-----------|----------|----------|----------------|-----------|----------|----------|
|                  | <i>b</i>           | <i>SE</i> | <i>z</i> | <i>p</i> | <i>b</i>       | <i>SE</i> | <i>z</i> | <i>p</i> |
| Exp. 1 (English) | 4.3                | 1.0       | 4.4      | < .001   | 4.5            | 1.0       | 4.5      | < .001   |
| Exp. 2 (English) | 2.9                | 0.8       | 3.7      | < .001   | 1.8            | 0.8       | 2.3      | .020     |
| Exp. 3 (English) | 3.0                | 0.4       | 7.1      | < .001   | 2.9            | 0.4       | 6.8      | < .001   |
| Exp. 4 (French)  | 1.0                | 0.4       | 2.7      | .006     | 0.8            | 0.4       | 2.1      | .036     |

- The same basic pattern in **all four** experiments:
  - **More literal** responses with **truth** than with felicity
  - **More literal** responses with **world knowledge** (WK) than with pictures
- But: effects are **smaller in French** (Exp. 4) than in English (Exps. 1–3)

Conclusion

Task design systematically influences how underinformative sentences are interpreted.

References

[1] W. De Neys and W. Schaeken. “When people are more logical under cognitive load: Dual task impact on scalar implicature”. In: *Experimental Psychology* 54.2 (2007), p. 128. [2] B. van Tiel, E. Pankratz, and C. Sun. “Scales and scalarity: Processing scalar inferences”. In: *Journal of Memory and Language* 105 (2019), pp. 93–107. [3] L. Bott and I. A. Noveck. “Some utterances are underinformative: The onset and time course of scalar inferences”. In: *Journal of Memory and Language* 51.3 (2004), pp. 437–457. [4] B. Van Tiel and W. Schaeken. “Processing conversational implicatures: Alternatives and counterfactual reasoning”. In: *Cognitive Science* 41 (2017), pp. 1119–1154. [5] A. Cremers and E. Chemla. “Direct and indirect scalar implicatures share the same processing signature”. In: *Pragmatics, semantics and the case of SIs*. Springer, 2014, pp. 201–227. [6] A. Feeney et al. “The story of some: Everyday pragmatic inference by children and adults”. In: *Canadian Journal of Experimental Psychology* 58.2 (2004), pp. 121–132. [7] J. Zehr and F. Schwarz. *PennController for Internet Based Experiments (IBEX)*. 2018. [8] A. M. Collins and M. R. Quillian. “Retrieval time from semantic memory”. In: *Journal of Verbal Learning and Verbal Behavior* 8.2 (1969), pp. 240–247. [9] H. H. Clark and W. G. Chase. “On the process of comparing sentences against pictures”. In: *Cognitive Psychology* 3.3 (1972), pp. 472–517. [10] M. Kissine and P. De Brabanter. “Pragmatic responses to under-informative some-statements are not scalar implicatures”. In: *Cognition* 237 (2023), p. 105463.